1. Design a counter with T flip flops whose count sequence is 0, 2, 4, 5, 3, 1, 0 . . .

Q_n			Q_{n+1}			T		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	1	0	0	1	0
0	1	0	1	0	0	1	1	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	1	1	1	0
0	1	1	0	0	1	0	1	0
0	0	1	0	0	0	0	0	1

T_2

$Q_2 \backslash Q_1Q_0$	00	01	11	10
0				1
1		1	X	X

$T_2 = Q_2 Q_0 + Q_1 \overline{Q}_0$

T_1

$Q_2 \backslash Q_1Q_0$	00	01	11	10
0	1		1	1
1		1	X	X

$T_1 = Q_1 + \overline{Q}_2 \overline{Q}_0 + Q_2 Q_0$

T_0

$Q_2 \backslash Q_1Q_0$	00	01	11	10
0		1		
1	1		X	X

$T_0 = Q_2 \overline{Q}_0 + \overline{Q}_2 \overline{Q}_1 Q_0$

2. Design a 3 bit up/down counter using negative edge triggered T-FF.

For up count, up=1

$Q_2Q_1Q_0$	$Q_2^+Q_1^+Q_0^+$	$T_2T_1T_0$
0 0 0	0 0 1	0 0 1
0 0 1	0 1 0	0 1 1
0 1 0	0 1 1	0 0 1
0 1 1	1 0 0	1 1 1
1 0 0	1 0 1	0 0 1
1 0 1	1 1 0	0 1 1
1 1 0	1 1 1	0 0 1
1 1 1	0 0 0	1 1 1

$T_0 = 1$

$T_1 = Q_0$

$T_2 = Q_0Q_1$

For down count, up=0

$Q_2Q_1Q_0$	$Q_2^+Q_1^+Q_0^+$	$T_2T_1T_0$
1 1 1	1 1 0	0 0 1
1 1 0	1 0 1	0 1 1
1 0 1	1 0 0	0 0 1
1 0 0	0 1 1	1 1 1
0 1 1	0 1 0	0 0 1
0 1 0	0 0 1	0 1 1
0 0 1	0 0 0	0 0 1
0 0 0	1 1 1	1 1 1

$T_0 = 1$

$T_1 = \overline{Q_0}$

$T_2 = \overline{Q_0}\,\overline{Q_1}$

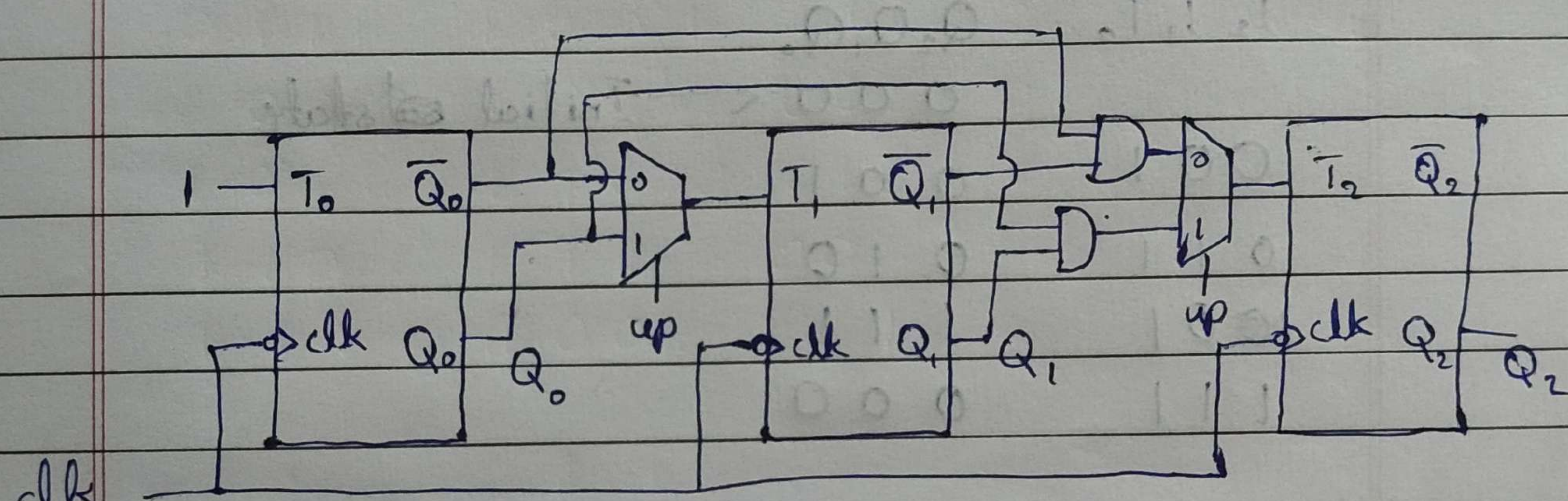
For up count up = 1
down count up = 0

$T = T_2 T_1 T_0$

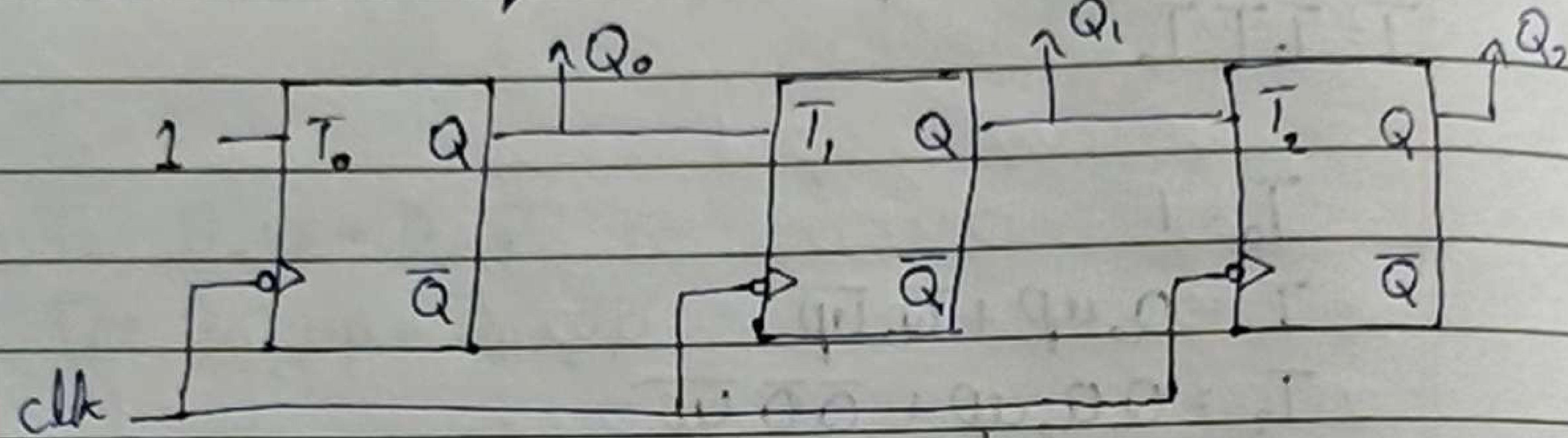
$T_0 = 1$

$T_1 = Q_0 \, up + \overline{Q_0} \, \overline{up}$

$T_2 = Q_1 Q_0 \, up + \overline{Q_1} \, \overline{Q_0} \, \overline{up}$



3. The circuit in the figure below is a counter. What is the sequence that this circuit counts in?



$\overline{Q_1}$ $\overline{Q_0}$ 1

$T_2\,T_1\,T_0$	$Q_2\,Q_1\,Q_0$	
	0 0 0	← Initial state
0 0 1	0 0 1	
0 1 1	0 1 0	
1 0 1	1 1 1	
1 1 1	0 0 0	

∴ the sequence is 000, 001, 010, 111, 000, ...